

B.E. Final Year VII Sem. 2016

(Design Engineering Quiz)

**Maximum Marks 50
minutes**

Time: 1 hour 30

Note: Attempt all Questions.

1. Design a 9 step gear box for the following Specification:

- Maximum Cutting Speed 55m/min.
- Minimum Cutting Speed 15m/min.
- Minimum diameter to be turned = 20mm.
- Range ratio should be less than 25

- (a.) Draw the possible structure diagrams and select the most optimum one with reason.
- (b.) Draw the Ray diagram with motor speed 720 rpm.
- (c.) Draw the Physical layout of the gear box
- (d.) Also calculate the number of teeth on gears and the variation in the calculated speed and the actual Speed Assume Suitable value of common ratio. (40)

- 2. Write down the procedure for selection of ball bearing for the above gear shafts. (10)**

DEPARTMENT OF MECHANICAL ENGINEERING
FACULTY OF ENGINEERING AND ARCHITECTURE
JAI NARAIN VYAS UNIVERSITY
B.E. FINAL YEAR VII SEMESTER
DESIGN ENGINEERING (QUIZ)

Max. Marks 50

Time 2 Hrs.

1. For a speed gear box having 6 steps with minimum speed of 45 rpm and maximum 250 rpm., The common ratio is 1.41. Determine the following.
 - (e) Draw all possible structure diagrams and select the optimum one with justification
 - (f) Draw the speed chart (Ray diagram)
 - (g) Draw the physical layout of the gear box.
 - (h) Determine the ratio of transmission at each stages and number of teeth on gears
(10×4=40)
 2. Write down the procedure to design a pair of spur gear for a given power and speed reduction ratio.
(10)
- OR
2. Write down the procedure to design an spindle of a lathe machine. Input required may be assumed.

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No.78538

FINAL B.E.

5

Opert. Res.

**FINAL B.E. (MECHANICAL ENGINEERING)
VII SEMESTER EXAMINATION-2017
ME-405A : OPERATIONS RESEARCH**

Time Allowed- Three Hours
Maximum Marks : 70

Note:- (1) Answer any FIVE questions.

(2) Marks allotted to each part of the question are indicated on the right side.

1. Define 'Operations research'. Explain characteristics, scope and application of OR in industry. 14

2. A company produces two type of toys labour time required for the first type is twice that of the second. If first type alone is produced the capacity is sufficient to produce 1000 toys per day. The demand potential is 300 and 500 days respectively for the two types. The profit per day is Rs 10 and Rs 8 respectively

(i) Formulate the LPP for maximizing profit.

(ii) Solve problem graphically

(iii) Make dual of the problem formulated. $5+5+4=14$

3. Solve the following LPP by simplex procedure

8332

1

(Contd.)

Minimise $z = 2x_1 + 5x_2$
 Subject to $x_1 + x_2 = 100$
 $x_1 \leq 40$
 $x_2 \geq 30$
 $x_1, x_2 \geq 0$

$$x_1, x_2 \geq 0$$

14

4. Availability of certain commodity in the warehouses and the requirements in various sales depots and cost of transporting a unit from the warehouse to depot are given as under:

WH \ Depot	S1	S2	S3	Availability
W1	-10	-20	-5	4
W2	-5	-10	-10	3
Demand	2	2	3	

Determine the best allocations for least transportation cost using VAM

5. A company is faced with the problem of assigning six different machines to five different jobs. One job on one machine. Cost in hundred of rupees are estimated as under.

Machine	A	B	C	D	E
I	2.5	5	1	6	1
II	2	5	1.5	7	3
III	3	6.5	2	8	3
IV	3.5	7	2	9	4.5
V	4	7	3	9	6
VI	6	9	5	10	6

Find optimum assignment & minimum total cost.

6. (a) Solve following game problem & value of game.

8332

2

(Contd.)

Player A	A ₁	A ₂	A ₃	A ₄
Player B	3	2	4	0
A ₁	3	4	2	4
A ₂	4	2	4	0
A ₃	0	4	0	8

7

(b) A retailer buys an item at Rs 2 per case and sells the same at Rs 5 per case. Any item not sold has no value. How much quantity of cases he must stock? Previous experience shows that demand varies from 13 to 15 cases a day.

(i) Prepare payoff table
 (ii) Suggest his decision as per maximax maximum & Laplace criterion.

(iii) Prepare regret (Opportunity loss table) table.

(iv) Find Expected value of Perfect information if probability of demand from 13 to 15 cases is 0.3, 0.5, 0.2 respectively each day.
 $2 + 1\frac{1}{2} + 1\frac{1}{2} + 2 = 7$

(a) Derive EOQ formula with finite production rate, shortage not permitted also give expression for total variable cost per year when we order in EOQ. (give assumption of the model). 8

(b) Write short notes on any two

- Components of queue problem
- Simulation technique & its application
- Slack, Surplus and artificial variables
- Infeasible, unbounded & Degenerate LP Problems. 3+3=6

8332

3

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No.15MECSB512

Final B.E.

5

Opert. Res.

**FINAL B.E. (MECHANICAL
ENGINEERING)
VII SEMESTER EXAMINATION-2018
ME-405 A: OPERATIONS RESEARCH**

Time Allowed - Three Hours
Maximum Marks - 70

- Note :-** (1) Answer any **FIVE** questions.
(2) Marks allotted to each part of the question are indicated on the right side.

- X
- 2- (a) Explain different OR models developed and applied to solve problem in business & industry. (14)
- The ABC Electrical Appliance. Company produces two products: refrigerators and ranges. Production takes place in two separate departments. Refrigerator are produced in deptt. I and ranges in dept II. The weekly production can not exceed 25 refrigerations in deptt I and 35 ranges in deptt II. The company regularly employs a total of 60 workers in the two deptts. A refrigerator requires 2 man weeks of labour while a range requires 1 man

week of labour. A refrigerator contributes a profit of Rs 600 and a range contribute a profit of Rs 400. Formulate a LP problem to determine number of units of both products to maximise weekly profit. Solve the problem graphically to determine optimal mix. (10)

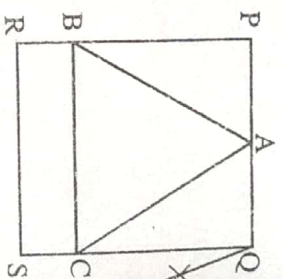
(b) Explain infeasible and unbounded solution in LP.

Write an LP for each case which will give infeasible solution and unbounded solution respectively. (4)

Food X contains 6 units of Vitamin A per gram and 7 units of vitamin B per gram and costs 12 paise per gram. Food Y contains 8 units of per gram A and 12 units of vitamin B per gram and costs 20 paise per gram. The daily minimum requirement of vitamin A and B is 100 units and 120 units respectively. Find optimal mix by simplex method. (10)

(b) Write dual of LP formulated in 3(a). (4)

A company has factories at A, B and C. Which supply warehouse at P, Q, R and S. Weekly factor capacities are 250, 300 and 400 units respectively for normal shift. Factory A and B are capable of overtime production of 50 and 75 units per week with an incremental cost of Rs 4 and Rs 5 respectively. The current warehouse requirements are 200, 275, 275 and 300 units respectively. The positions of plant and warehouses are as shown below.



Roads PQ, QS and RP form a square of 10 km side, where as roads AB, BC and CA form an equilateral triangle. A lies midway between P and Q. Transportation cost is estimated at Rs 2.50 per km. Formulate above as a transportation model and give initial feasible solution by VAM. (Calculate distance upto 2 decimal places.)

A manufacturer of complex electronic equipment has just received a sizable contract and plans to subcontract part of job. He has solicited bids for 6 subcontracts from 3 firms. Each job is sufficiently large and any firm. Can take only one job. The table below shows the bids, as well as the cost estimate (in lakhs of rupees) for doing the job internally. Maximum three jobs can be performed internally. Find the optimal assignment that will result in minimum total cost. (14)

An icecream retailer buys icecream at a cost of Rs 5 per cup and sells it for Rs 8 per cup, any remaining unsold at the end of the day can be disposed at a salvage price of Rs 2 per cup. Past sales have ranged between 15 and 18 cups per day with the given probabilities as under.

Sales	15	16	17	18
(Cups/day) Probability	0.10	0.20	0.40	0.30

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No. ~~MA~~...16MEC38511

FINAL B.E.

5

Opert. Res.

FINAL B.E. (MECHANICAL ENGINEERING)

VII SEMESTER EXAMINATION - 2019

ME - 405 A: OPERATIONS RESEARCH

Time - Three Hours

Maximum Marks - 100

Note:- (1) Answer any **Five** questions.

(2) Marks allotted to each part of the question are indicated on the right side.

①

(a) Use the graphical method to solve the following Linear Programming problem:

$$\text{Maximize } Z = 2x_1 + x_2$$

subject to the constraints

$$(i) \quad x_1 + 2x_2 \leq 10$$

$$(ii) \quad x_1 + x_2 \leq 6$$

$$(iii) x_1 - x_2 \leq 2$$

$$(iv) x_1 - 2x_2 \leq 1$$

$$x_1, x_2 \geq 0$$

(b)

An Air Force is experimenting with three types of bombs P, Q and R, in which three kinds of explosives, viz., A, B and C will be used. Taking the various factors into account, it has been decided to use the maximum 600 Kg of explosive A, at least 480 Kg of explosive B and exactly 540 Kg of explosive C. Bomb P requires 3.2, 2.2 Kg, bomb Q requires 1.4, 3 Kg and bomb R requires 4.2, 3 Kg of explosives A, B and C respectively. Bomb P is estimated to give the equivalent of a 2 ton explosion, bomb Q, a 3 ton explosion and bomb R a 4 ton explosion respectively. Under what production schedule can the Air force make the biggest bang?

Only formulate the linear programming problem.

(c) Explain the conditions for following in L.P.

(i) Unbounded solution

(ii) Infeasible solution

10+5+5

For a linear programming Problem (Q1(b)), the partial solution is given below in a table. Fill the 'question mark' places and final the optimal solution. S_1, S_2, A_1 are

3967

2

(Contd.)

slack, surplus and artificial variables.

$C_j \rightarrow$		2	3	4	0	0	0	-M	
Basic variable coefficients (C_b)	Basic variable value (b)								Minimum Ratio
?	?	x_1	x_2	x_3	S_1	S_2	A_1		960/7
?	480	5/2	0	7/2	1	1/4	0		240
?	120	1/2	1	1/2	0	-1/4	0		
?	?	1/3	0	3/2	0	3/4	1		120
$Z = 360 - 180M$									
$Z_j - Z_j \rightarrow$		3/2 - M/2	?	3/2 - 3M/2	0	-3/4 - 3M/4	-M		20

3.

A company has three production facilities S_1, S_2 and S_3 with production capacity of 7, 9 and 18 units per week of a product, respectively. These units are to be shipped to four warehouses D_1, D_2, D_3 and D_4 with requirements of 5, 8, 7 and 14 units per week, respectively. The transportation costs (in rupees) per unit between factories to warehouses are given in the table below:

	D_1	D_2	D_3	D_4
S_1	19	30	50	10
S_2	70	30	40	60
S_3	40	8	70	20

Formulate this transportation problem & solve to minimize the total transportation cost. 20

4. (a) In the modification of a plant layout of a factory four new machines M_1, M_2, M_3 and M_4 are to be installed in a machine shop. There are five vacant places A, B, C, D and E available. Because of limited space, Machine M_2

3967

3

(Contd.)

cannot be placed at C and M_3 cannot be placed at A. The cost of locating a machine at a place is as follows

Machines	Locations				
	A	B	C	D	E
M_1	9	11	15	10	11
M_2	12	9	—	10	9
M_3	—	11	14	11	7
M_4	14	8	12	7	8

Find the optimal assignment schedule.

(b) Two firms A and B have, for years, been selling a competing product that forms a part of both firms' total sales. The marketing executive of firm A raised the question: 'What will be the firm's percentage gain in terms of advertising for the product in question? The market research team of firm A developed the following data for varying degree of advertising:

- No advertising, medium advertising and large advertising for both firms will result in equal market shares.
- Firm A market share with no advertising : 40 percent of the market with medium advertising by firm B and 28 percent of the market share with large advertising by firm B.
- Market share of Firm A using medium advertising 70 percent of the market with no advertising by the firm B and 45 percent of the market with large advertising by firm B.

3967

4

(Contd.)

- Market share of firm A using large advertising : 75 percent of the market with No advertising by firm B and 47.5 percent of the market with medium advertising by firm B.

Based on the above information what will be the answer to the marketing executive's question.

10+10

3. (a)

A food product's company is contemplating the introduction of a revolutionary new product with new packing as replacing the existing product at much higher price (S_1). It may even make a moderate change in the composition of existing product, with a new packaging at a small increase in price (S_2), or may put a small change in the composition of the existing product, backing it with the word 'New' at a negligible increase in price (S_3). The three possible states of nature are : (i) high increase in sales (N_1), (ii) No change in sales (N_2) and (iii) decrease in sales (N_3). The marketing department of the company worked out the payoffs in terms of yearly profit for each of the decisions of three natures (expected sales). This is represented in the following table:

Decisions	State of Nature		
	N_1	N_2	N_3
S_1	7,00,000	3,00,000	1,50,000
S_2	5,00,000	4,50,000	0
S_3	3,00,000	3,00,000	3,00,000

5

(Contd.)

3967

Which decision should the concerned executive choose on the basis of:

- (i) Maxi - min Criterion
- (ii) Maxi - max Criterion

(b) The following information is provided for an item:

Annual demand : 12,000 units;

Ordering cost : Rs 60 per order; Carrying cost : 10%,
Unit cost of item : Rs 10 and Lead time : 10 days.

There are 300 working days in a year. Determine EOQ and number of orders per year. In the past two years the demand rate has gone as high as 70 units per day. For a reordering system, based on the inventory level, determine buffer stock. 10+10

6. An established company has decided to add a new product to its line. The steps shown below in the table are to be planned.

Activity	Description	Predecessors	Duration (days)
A	Organize sales office	—	6
B	Hire salesmen	A	4
C	Train Salesmen	B	7
D	Select advertising agency	A	2
E	Plan advertising campaign	D	4
F	Conduct advertising campaign	E	10

3967

6

(Contd.)

G	Design package	—	2
H	Setup packaging facilities	G	10
I	Package initial stocks	J, H	6
J	Order stock from Manufacturer	—	13
K	Select distributors	A	9
L	Sell to distributors	C, K	3
M	Ship stocks to distributors	L, L	5

- (a) Draw network diagram
- (b) Indicate the critical path. 20

7. (a) Arrivals at telephone booth are considered to be Poisson with an average time of 10 minutes between on arrival and the next. The length of phone calls is assumed to be distributed exponentially, with a mean of 3 minutes.

- (i) What is the probability that a person arriving at the booth will have to wait?
 - (ii) The telephone department will install a second booth when convinced that an arrival would expect waiting of at least 3 minutes for a phone call. By how much should the flow of arrivals increase in order to justify second booth?
 - (iii) What is the average length of the queue that forms from time to time?
- (b) Write a short notes on any TWO of following:
- (i) Methods to find Basic Feasible solutions in Transportation problem.

3967

7

(Contd.)

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No. ...7...2...539...

Final B.E.

4

Manufact. Tech.

FINAL B.E.

(MECHANICAL ENGG.)

VII SEMESTER EXAMINATION - 2017

ME-404 A : MANUFACTURING TECHNOLOGY

Time - Three Hours

Maximum Marks : 70

Note :- (1) Answer any Five questions.

(2) Marks allotted to each part of the question are indicated on the right side.

- X
1. (a) If σ_1 , σ_2 and σ_3 are principal stresses ($\sigma_1 > \sigma_2 > \sigma_3$), find the expression for yield criteria for plastic deformation in metal working according to Tresca and von-Mises and compare the two. 7
 - (b) Explain various types of function and their effect in metal working. Discuss the role of lubrication in metal forming. 7
 2. (a) What are the various types of forging? Explain the process and various passes in drop forging. 7
 - (b) Briefly discuss the following sheet metal working processes- shearing, blanking, punching and bending. 7

3. (a) With the help of neat sketches, discuss the effect of following parameters on metal removal rate in abrasive jet machining:

- Abrasive mass flow rate
- Velocity of particles
- Nozzle tip distance

7

- (b) Explain the construction and working principle of plasma arc machining. Discuss its advantages, limitations and applications.

7

4. (a) What are the various methods of manufacturing external threads? Describe thread grinding process with the help of neat sketch.

7

- (b) Differentiate between gear shaping and gear shaving processes.

7

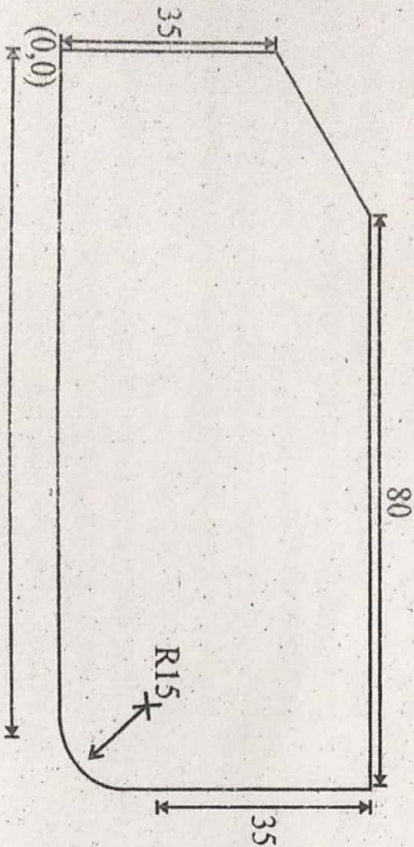
5. (a) What is rapid prototyping (RPT)? Explain various techniques used for RPT. Discuss the advantages and applications of RPT.

7

- (b) Describe various configurations of robot with the help of neat sketches.

7

6. Write a manual part program using G and M codes for preparing the job shown in the diagram below.



8331

95
2

(Contd.)

The job is to be produced on a CNC milling machine using a cutter. (tool no. 2) having diameter 12mm (cutter radius compensation is stored in memory address D 02), at a speed of 1000 rpm in clockwise direction. The feed rate is 65mm/min and depth of cut is 10mm. The home position of the tool is at point (-20,0,20). The top surface of the workpiece is at $Z=0$

14

7. Write short notes on any four :-

- Principal stress and strain
- High energy rate forming
- Chemical milling
- Gear stamping
- End effectors
- Flexible manufacturing system

14

8331

3

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No. 15ME58512

FINAL B.E.

4

Manufact. Tech.

**FINAL B.E. (MECHANICAL ENGG.)
VII SEMESTER EXAMINATION 2018
ME-404 A : MANUFACTURING
TECHNOLOGY**

Time - Three Hours
Maximum Marks - 70

- Note :-** (1) Attempt any **FIVE** Questions.
(2) Marks allotted to each part of the question are indicated on the right side.

1. (a) Electrochemical Machining is performed to remove material from an Iron surface of 20mm x 20mm under the following conditions :
Inter Electrode gap = 0.2 mm
Supply voltage (DC) = 12 V
specific resistance of electrolyte = $2 \Omega \text{ cm}$
Atomic weight of Iron = 55.85
Valency of Iron = 2
Faraday's Constant = 96540 coulombs.
What is the material removal rate in (gm/S) (7)

(b) Explain the construction and working principle of electrode discharge machining? Draw the variation curve of current and voltage and the applications of EDM? (7)

2. (a) Explain the various types of Defects in metal forming? What are the advantages and disadvantages of Hot and Cold forming? (7)

(b) An Aluminium rod of 6.25 mm diameter is drawn in to wire of 5.6 mm diameter. Semi die angle, considering the friction to be 0.04 and nominal stress 35MPa. also calculate the maximum reduction that can be given to material. (7)

3. (a) Find the expression for yield criteria for plastic deformation in metal working according to Tresca's maximum shear stress criterion? Discuss the role of lubrication in Metal forming? (7)

(b) Explain the term extrusion? What are the various types of Extrusion? Explain the process of any two type of extrusion? (7)

4. (a) What are the various methods of manufacturing gears? Explain them briefly. What are the gear finishing methods. (7)

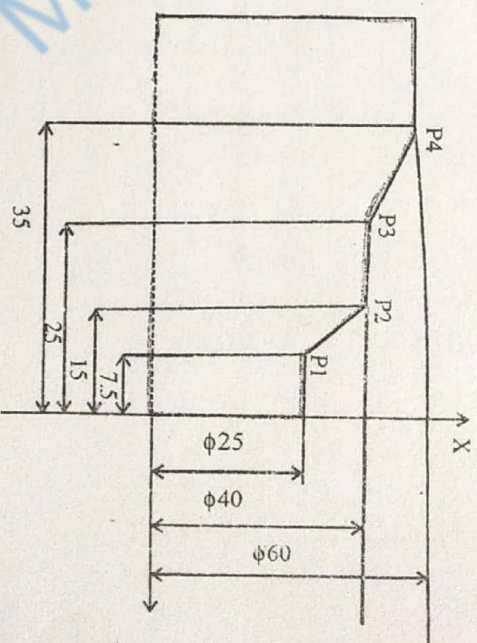
(b) Briefly discuss the following sheet metal working processes. (7)

- (i) High Energy rate forming
- (ii) Drawing
- (iii) Punching
- (iv) Spinning

5. (a) Describe the configurations of mechatronics with the help of neat sketches? (7)

(b) what is rapid prototyping (RPT)? Explain various techniques used for RPT. What are the key benefits of using Rapid Prototyping for product design & Development? (7)

6. Write a manual part programming using G and M codes for preparing the job shown in the diagram below :- (14)



Take all the assumptions which is required for programming.

FINAL B.E.

Roll No. 16MEC38511

4

Manufact. Tech.

FINAL B.E. (MECHANICAL ENGINEERING)
VII SEMESTER EXAMINATION - 2019
ME - 404 A: MANUFACTURING TECHNOLOGY

Time - Three Hours

Maximum Marks - 100

Note:- (1) Answer any FIVE questions.

(2) Marks allotted to each part of the question are indicated on the right side.

1. (a) Considering plane stress system, find magnitudes and directions of principal stresses, maximum shearing stress and corresponding normal stress. Also construct Mohr's circle for

$$\sigma_x = 750 \text{ N/mm}^2; \sigma_y = 150 \text{ N/mm}^2; \sigma_z = 0 \text{ and } \tau_{xy} = 150 \text{ N/mm}^2$$

12

(b) For problems given in question 1(a), find yield stress of

the material according to :

(a) Tresca criteria

(b) Von - Mises Criteria

8

2(a) Differentiate between the various forging processes and derive expression for maximum force required for forging. 10

(b) A hot upset forging operation is performed in an open die. The initial size of the work part is : $D_0 = 25\text{mm}$, and $h_0 = 50\text{mm}$. The part is upset to a diameter = 50mm . The work metal at this elevated temperature yields at 85MPa ($n=0$). Coefficient of friction at the die-work interface = 0.40 . Determine (a) final height of the part, and (b) maximum force in the operation. 10

3. (a) Explain rolling process and classify it. Derive expression for roll separating force, driving torque and power in rolling operation. 10

(b) A plate that is 250mm wide and 25mm thick is to be reduced in a single pass in a two-high rolling mill to a thickness of 20mm . The roll has a radius = 500mm , and its speed = 30m/min . The work material has a strength coefficient = 240MPa and a strain hardening exponent = 0.2 . Determine (a) roll force, (b) roll torque, and (c) power required to accomplish this operation. 10

4. (a) Highlight the importance of extrusion process and derive expression for frictional power loss and maximum force required in extrusion process. 10

3966

2

(Contd.)

(b) A 3.0-in - long cylindrical billet whose diameter = 1.5in is reduced by indirect extrusion to a diameter = 0.375in . Die angle = 90° . In the Johnson equation, $a = 0.8$ and $b = 1.5$. In the flow curve for the work metal, $K = 75,000\text{lb/in}^2$ and $n = 0.25$. Determine (a) extrusion ratio, (b) true strain (homogeneous deformation), (c) extrusion strain, (d) ram pressure, (e) ram force, and (f) power if the ram speed = 20in/min . 10

5(a) Compare Electrochemical Machining and Electric Discharge Machining in terms of operating principle, key elements and process parameters. 12

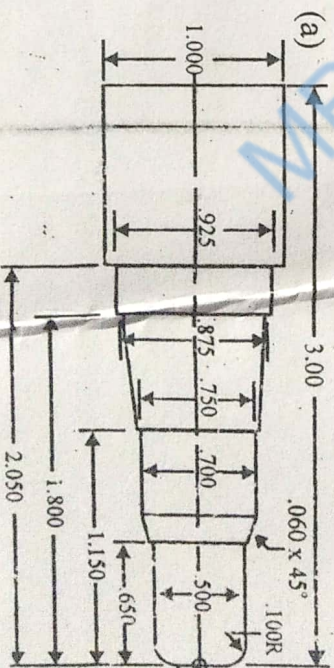
(b) Explain Plasma Arc Machining in details. Discuss about the effect of its process parameters. 8

6 (a) Describe various methods of producing screw threads: Casting, Forming, Machining and Grinding. 10

(b) Classify the gears and write about their applications. Give detailed explanation of the specifications of a gear. 10

7. Write CNC part program for following:

10



3

(Contd.)

3966

Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No. 16MEC38511

Final B.E. (M)

3

Hydraulic Machines

FINAL B.E. (MECHANICAL ENGINEERING)

VII SEMESTER EXAMINATION - 2019

ME- 403 A : HYDRAULIC MACHINES (M)

Time - Three Hours

Maximum Marks - 100

Note:- (1) Attempt any FIVE questions.

(2) Marks allotted to each part of the question are indicated on the right side.

1. (a) A symmetrical vane moves with a velocity u , cross-sectional area of jet 'a' and velocity v strikes the vane at center and in the direction of motion. Prove that for maximum efficiency of vane,

$$\eta_{\max} = \frac{8}{27} (1 + \cos \theta); \text{ where } \theta \text{ is vane tip angle. } 10$$

- (b) A lawn sprinkler with two nozzles of 5 mm diameters each at 20 cm and 15 cm radii is

connected to a tap discharges 6 liters/minute. The nozzle discharge water upwards to the plane of rotation. What torque will the sprinkler exert on hand, if (i) held stationary and (ii) at what angular velocity will it rotate freely? 10

2. (a) Water jet of velocity 20 m/s impinges on a curved vane deflects jet through 120° having vane angle $(\theta \text{ and } \phi \text{ each})$ of 30° . If vane is moving at 5 m/s, find (i) vane angle to avoid shock (ii) absolute exit velocity in magnitude and direction and (iii) the work done. 10

- (b) A boat travelling at 12 m/s in fresh water has 0.6 m diameter propeller which takes 4.25 m^3 of water per second between blades. Assuming the effect of propeller hub and boat hull on flow are negligible, calculate (i) thrust on the boat, (ii) theoretical efficiency of the propulsion and (iii) power input to the propeller. 10

- (3) (a) Why is the casing of reaction turbines are of spiral shape with uniform change in cross sectional area? In an inward flow reaction turbine the speed ratio is k_u , flow ratio is ψ and breadth ratio is n . If the overall efficiency of turbine is 85% and area ratio of vanes at inlet is 5% of peripheral area at inlet then show that $N_s = 888.23 k_u (\psi n)$. 10

(Contd.)

2

3965

- (b) An inward flow reaction turbine has a runner of 60 cm diameter and breadth at inlet is 6 cm. The inner diameter is 45 cm and breadth at outlet is 8 cm. The vanes occupy 5% of peripheral area. The guide vane angle at inlet is 22° and vane angles at inlet and outlet are 80° and 25° respectively. The turbine head is 60 m and hydraulic efficiency is 90%. Assuming the mechanical efficiency of 95%, calculate (i) the power delivered to the turbine also determine (ii) the rotational speed of the turbine. 10

4. (a) How is the maximum pressure controlled in a variable volume vane-type pump? Prove, the work saved by fitting air vessels in a double acting pump is about 39.2%. 10

- (b) Determine the diameters and blade angles of a Francis turbine running at 500 rpm under a head of 120 m and delivering 3 MW power. Assume, flow ratio as 0.14 and $D_2 = 0.5 D_1$ and $b_1 = 0.1 D_1$ with usual notations. The hydraulic efficiency is 90% and overall efficiency is 84%. 10

5. (a) What is the basic operating principle of a hydraulic drive? List advantages of fluid drive. Explain the basic operating principle of hydraulic motor in hydraulic circuit. 10

(Contd.)

3

3965

(b) A Kaplan turbine develops 2 MW under a head of 6 m. The turbine is set at 2.5 m above the tail race level. A vacuum gauge inserted at the turbine outlet records a suction head of 3.1 m. If the turbine efficiency is 85% what would be the (i) efficiency of the draft tube having inlet diameter of 3 m? Also find the reading of the (ii) suction gauge if power developed is reduced to 1000 kW, the head and speed remaining constant. 10

(a) The inlet of a draft tube for a reaction turbine is 2.5 m above the tail race level. The outlet area is 3 times the inlet area, the velocity at inlet is 8 m/s, and kinetic head recovery is 80%. Considering atmospheric head as 10 m water column, determine pressure at draft tube inlet. 10

(b) A centrifugal pump is designed to run at 950 rpm to deliver $0.4 \text{ m}^3/\text{s}$ to a head of 16 m. If the pump is coupled to a motor of speed 1450 rpm. Calculate the discharge, head and power input. Assume, overall efficiency is 0.82 remains constant. 10

(a) Name three moving elements necessary for operation of a reciprocating pump. How is capacity of a pump calculated? List four strokes of a single-acting lift pump. 10

(b) It is desired to discharge water at 10 litre/min by a reciprocating pump have bore to stroke ratio of

1:1.5 and running at 42 rpm. The expected slip is 12%. Determine the bore and stroke for a (i) single acting pump, and (ii) double acting pump. If the total head is 30m and overall efficiency is 82%, find (iii) the power required in both cases. 10

(Contd.)

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No. 14MEC 78521

Final B.E.

1

Heat & Mass Transfer

FINAL B.E. (MECHANICAL ENGINEERING)

VII SEMESTER EXAMINATION - 2017

ME-401 A : HEAT & MASS TRANSFER-I

Time - Three Hours

Maximum Marks : 70

Note : - (1) Attempt FIVE questions.

(2) Marks allotted to each part of the question are indicated on the right side.

(3) Use of steam tables and other property tables is permitted.

(4) Assume any missing data and mention it clearly.

1. (a) Derive a general heat conduction equation in cylindrical coordinates for three dimensional unsteady heat conduction by considering infinitesimal volume element. 8

1. (b) What are boundary and initial conditions? Describe different types of boundary conditions applied to heat conduction problem. 6

2. (a) Derive an expression for heat conduction through a sphere and prove that if the thickness of the sphere is small it can be taken as a flat slab. 7

2.

- (b) A 30 cm thick wall of 5m x 3m size is made of red brick ($k=0.3 \text{ W/m-deg}$). It is covered on both sides by layers of plaster, 2 cm thick ($k=0.6 \text{ W/m-deg}$). The wall has a window side of 1m x 2m. The window door is made of 12 mm thick glass ($k=1.2 \text{ W/m-deg}$). if the inner and outer surface temperatures are 15 and 40°C, make calculations for the rate of heat flow through the wall. 7

- (a) Show that in a long cylinder of radius R with uniformly distributed heat sources, the temperature distribution is prescribed by the relation. 7

$$\frac{t - t_w}{t_{\max} - t_w} = \left(\frac{r}{R} \right)^2$$

where t_w is the temperature at the outer surface of the cylinder and $t_{\max}$ is temperature along the cylinder axis. 7

- (b) For a constant cross-section area fin, obtain the temperature distribution and total heat flow rate under steady state conditions when one end of the fin is attached to a body at high temperature and other end of the fin is insulated. 7

4.

- (a) An egg with mean diameter of 4 cm and initially at 25°C is placed in a boiling water pan for 4 minutes and found to be boiled to the consumer's taste. For how long should a similar egg for same consumer be boiled when taken from a refrigerator at 5°C? Use lumped parameter theory and presume the following properties for egg: 7

$$k = 12 \text{ W/m-deg}$$

$$h = 125 \text{ W/m}^2\text{-deg}$$

$$C_p = 2 \text{ kJ/kg-K}$$

(Contd.)

8329

4.

and $\rho = 1250 \text{ kg/m}^3$ 7

- (b) A steel rod ($k=30 \text{ W/m-deg}$) 1 cm in diameter and 5 cm long protrudes from a wall which is maintained at 100°C, the rod is insulated at its tip and is exposed to an environment with $h=50 \text{ W/m}^2\text{-deg}$ and $t_\infty = 30^\circ\text{C}$. Calculate the fin efficiency, temperature at the tip of the fin and the rate of heat dissipation. 7

3.

- (a) State and explain the following laws relating to thermal radiation and temperature of a radiating body: 7

Planck's law, Stefan Boltzman law and Wien's displacement law 7

5.

- (b) Two large parallel planes with emissivity 0.4 are maintained at different temperatures and exchange heat only by radiation. What percentage change in net radiative heat transfer would occur if two equally large radiation shields with surface emissivity 0.04 are introduced in parallel to the plates? 7

6.

- (a) Using the definition of radiosity and irradiation, prove that the radiant interchange between two gray bodies is given by relation 7

$$Q_{net} = \frac{A\sigma_b(T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{1}{F_{12}} + \left(\frac{1 - \epsilon_2}{\epsilon_2} \right) \frac{A_1}{A_2}}$$

the notations have their usual meanings. 8

- (b) Calculate the shape factors for the configurations show in the figure given below 8

3

(Contd.)

8329

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No. ...SMEECS8512...

Final B.E.

1

Heat & Mass Transfer

**FINAL B.E.
(MECHANICAL ENGINEERING)
VII SEMESTER EXAMINATION-2018
ME-401 A : HEAT & MASS TRANSFER-I**

Time - Three Hours
Maximum Marks - 70

- Note :-
- (1) Attempt any **FIVE** Questions.
 - (2) Marks allotted to each part of the question are indicated on the right side.
 - (3) Use of steam tables and other property tables is permitted.
 - (4) Assume any missing data and mention it clearly.

1. (a)

What is thermal Diffusivity, explain its significance? Describe different types of boundary conditions applied to heat conduction problem. (6)

(b) Derive a general heat conduction equation in spherical coordinate for three dimensional unsteady heat conduction by considering infinitesimal volume element. (8)

2. (a) A plane brick wall 5m long x 3m high x 250 mm thick has temperature of 800°C and 20°C maintained on its bounding surfaces. Presuming that thermal conductivity of the brick material is related to its temperature in degree celsius by the relation :
 $K = 1.0(1 + 0.001t)$ Kcal/m-hr-deg make calculations for the average thermal conductivity, thermal resistance and heat loss from the wall. What would be temperature at 100mm distance from the wall surface at 800°C? (7)

(b) Derive an expression for heat conduction through a cylindrical wall by starting with general heat conduction equation in cylindrical coordinates. (7)

3. (a) Explain the critical thickness of insulation and prove that heat flow rate is maximum at $r_c = k / \lambda_0$ where r_c = critical radius, k = thermal conductivity and λ_0 = outside convective film coefficient. (7)

(b) Obtain the expression of temperature distribution and total heat flow rate under steady state conditions when one end of the fin is attached to a body at high temperature and other end is losing heat at the tip. (7)

4. (a) A 12 mm diameter mild steel sphere ($K=42.5$ W/m-K) is exposed to cooling air flow at 27°C resulting in the convective coefficient $h=114$ W/m²K. Determine time required to cool the sphere from 540°C to 95°C using lumped parameter theory. Given :
 $\rho_{steel} = 7850$ Kg / m³, $C_p = 475$ J / KgK
 thermal diffusivity $\alpha = 0.043$ m² / hr. Justify the value of Biot number for the analysis. (7)

(b) A very long copper rod 20mm in diameter extends horizontally from a plane heated wall maintained at 100°C. The surface of the rod is exposed to an air environment at 20°C with convective heat transfer coefficient of 8.5 W/m²-deg. Work out the heat loss if the thermal conductivity of copper is 400 W/m-deg. further estimate how long the rod be in order to be considered infinite. (7)

5. (a) State and explain the following laws relating to thermal radiation & temperature of a radiating body : Wein's displacement law Planck's law and Kirchhoff's law. (7)

(b) Two opposed, parallel, infinite planes are maintained at 420 K and 480 K respectively. Calculate the net heat flux between these planes if one has an emissivity of 0.8 and other an emissivity of 0.7. Does it matter which plate has which emissivity? How heat flux will be affected if the temperature difference is doubled by raising the temperature 480K to 540K? (7)

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No. 16 ME-C 38511

Final B.E. MECH.

1

Heat & Mass Transfer

FINAL B.E. (MECHANICAL ENGINEERING)

VII SEMESTER EXAMINATION - 2019

ME-401 A: HEAT & MASS TRANSFER - I

Time - Three Hours

Maximum Marks - 100

Note:- (1) Attempt any FIVE questions.

(2) Marks allotted to each part of the question are indicated on the right side.

(3) Use of steam tables and other property tables is permitted.

(4) Assume any missing data and mention it clearly.

Q

(a) Explain the mechanism of thermal conduction in gases, liquids and solids. Discuss the effect of temperature on thermal conductivity.

8

(Contd.)

(b) Derive a 3 - dimensional general conduction equation in cylindrical co-ordinates for a homogeneous material. Deduce therefrom an expression for unidirectional unsteady state system when heat is generated at uniform rate within the material. 12

(a) What is meant by overall heat transfer coefficient. Drive an expression for overall heat transfer coefficient (based on outer area) for the transfer of heat from a hot fluid flowing in a pipe to a cold fluid to which the pipe is exposed in a terms of film coefficient and radii of the pipe. 8

(b) A furnace wall is a made up of steel plate 10mm thick ($K = 62.8 \text{ kJ/m-hr-deg}$) lined on inside with Silica bricks 150 mm thick ($K = 7.32 \text{ kJ/m-hr-deg}$) and on the outside with magnesia bricks 200mm thick ($K = 18.84 \text{ kJ/m-hr-deg}$). The inside and outside surfaces of the wall are at temperature 650°C and 125°C respectively. Make calculations for the heat loss from unit area of the wall.

It is required that the heat loss be reduced to 10 MJ/hour by means of air gap between steel and magnesia bricks. Estimate the necessary width of air gap if thermal conductivity of air is $0.126 \text{ kJ/m-hr-deg}$. 12

3.

(a) Consider a long solid cylinder with uniform heat generation and with its outer surface maintained at a constant temperature. Set up an expression which describes the temperature distribution and specify the location of maximum temperature. 8

(b) A plate fin of 10mm thickness and 80mm length is dissipating heat from a surface at 190°C . The fin is exposed to air at 25°C with a convection coefficient of $22 \text{ W/m}^2 - \text{deg}$. If thermal conductivity of the fin material is 200 W/m-deg .

(a) determine the heat dissipation consider 1m width of fin.

(b) To increase the heat dissipation, the following two alternatives have been suggested with the same material volume.

(i) Split the fin into two fins of 5mm thickness each

(ii) Single fin 5mm thick and 160 mm long. Which will be the better choice?

The fin may be considered short with tip insulated. 12

(a) Discuss the criteria of selection of fins. What is the difference between the fin effectiveness and fin efficiency? 6

(b) Define Biot and Fourier numbers, and point out their physical significance. 4

(c) During a heat treatment process, alloy steel spherical ball of 12mm diameter are initially heated to 800°C in a furnace. Subsequently these are cooled to 100°C by keeping them immersed in an oil bath at 35°C with convection coefficient $20\text{W/m}^2\text{-deg}$. Determine the time required for the cooling process. Proceed to calculate the value of convection coefficient if it is desired to complete the cooling process in a period of 10 minutes. The thermo - physical properties of steel balls are:

Density = 7750 kg/m^3 ; Specific heat = 520 J/kgk
and conductivity = 50 W/m-deg . 10

5. (a) Making use of Plank's law of distribution establish the relation for the wien's displacement law. 8

(b) Calculate the shape factors for the configurations shown in the figure given below:

(i) Sphere of diameter d inside a cubical box of length $l = d$.

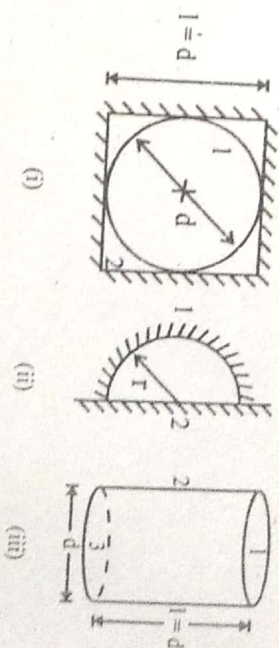
(ii) Hemispherical surface closed by a plane surface

3963

4

(Contd.)

(iii) End and side of circular tube of equal length and diameter. [take F_{13} as 0.17] 12



(a) Explain the "Surface resistance" and "Space resistance". How can you construct a radiation network for two gray surfaces exchanging radiant energy? 8

(b) Two large parallel planes with emissivity 0.4 are maintained at different temperatures and exchange heat only by radiation. What percentage change in net radiative heat transfer would occur if two equally large radiation shields with surface emissivity 0.04 are introduced in parallel to the plates? 12

3963

5

6. (a)

Define Radiosity and irradiation and Derive the relation for net heat exchange between two gray surface using electrical network approach. (7)

(b)

Consider radiative heat transfer between two large parallel planes of surface emissivities 0.8. How many thin radiation shields of emissivity 0.05 be placed between the surfaces to reduce the radiation heat transfer by a factor of 75 ? (7)

MBME-LEARNING